RAGLab AI

Experimental Evaluation Report

Comparative Analysis of Chunking Strategies and Embedding Models for RAG Retrieval

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Benchmark run ID: 4a7b2bb4-70e1-4100-bdfc-b200b94843d1

Dataset ID: 35ea593c-637b-42e7-810b-96647a56a617

Run status: Completed — 18/18 configurations; 72/72 measurements; 0 failures

This report documents an exploratory retrieval benchmark comparing three chunking strategies, two embedding models and three chunk-size settings. It summarizes the recorded measurements, configuration rankings, methodology and limitations.

1. Experimental Design

The experiment compared Fixed, Sentence and Semantic chunking strategies using all-MiniLM-L6-v2 and bge-small-en-v1.5 embedding models. Each strategy and model combination was evaluated at chunk sizes of 64, 128 and 256 words, giving 18 configurations in total.

Experiment dimension	Values
Chunking strategies	Fixed, Sentence, Semantic
Embedding models	all-MiniLM-L6-v2, bge-small-en-v1.5
Chunk sizes	64, 128, 256 words
Total configurations	18
Top-k	3
Repetitions	1
Validation questions	4 (3 answerable, 1 unanswerable)

2. Dataset and Evaluation Scope

The dataset contains 12 questions: four training, four validation and four test questions. Ten are answerable and two are unanswerable. All 12 questions were manually reviewed, with zero rejected. This benchmark used only the four validation questions, of which three were answerable.

Important: The source document is a fictional sample campus document. It does not represent actual PES University policies. The results are exploratory and should not be generalized to real institutional documents.

3. Evaluation Metrics

Metric	Meaning
Precision@3	Relevant retrieved passages among the three requested results.
Recall@3	Fraction of distinct validated evidence spans covered in the top three.
MRR	Reciprocal rank of the first relevant passage, averaged over answerable questions.
Retrieval latency	Measured query embedding, index validation/load and search time.

Metric	Meaning
Indexing time	Time recorded to build the retrieval index.
Abstention rate	Separate measure for declining retrieval on unanswerable questions.

4. Aggregate Results

The following tables report simple averages across the configurations in each group. They are descriptive summaries, not statistical significance tests.

4.1 Results by embedding model

Model	Precision@3	Recall@3	MRR	Latency (ms)	Indexing (ms)
bge-small	0.3457	1.0000	1.0000	83.64	108.33
minilm	0.3457	1.0000	1.0000	79.12	89.11

4.2 Results by chunking strategy

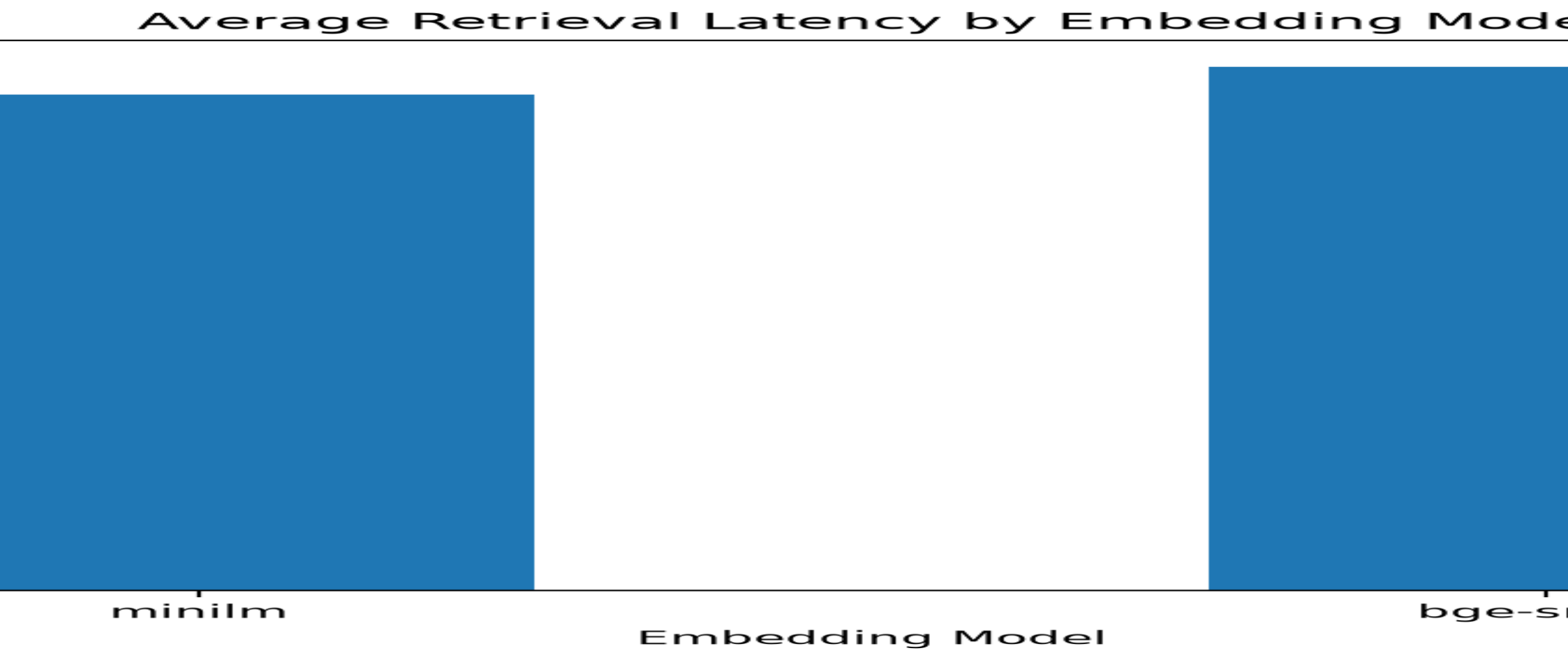
Strategy	Precision@3	Recall@3	MRR	Latency (ms)	Indexing (ms)
fixed	0.3704	1.0000	1.0000	81.29	74.09
semantic	0.3333	1.0000	1.0000	82.82	147.28
sentence	0.3333	1.0000	1.0000	80.04	74.78

4.3 Results by chunk size

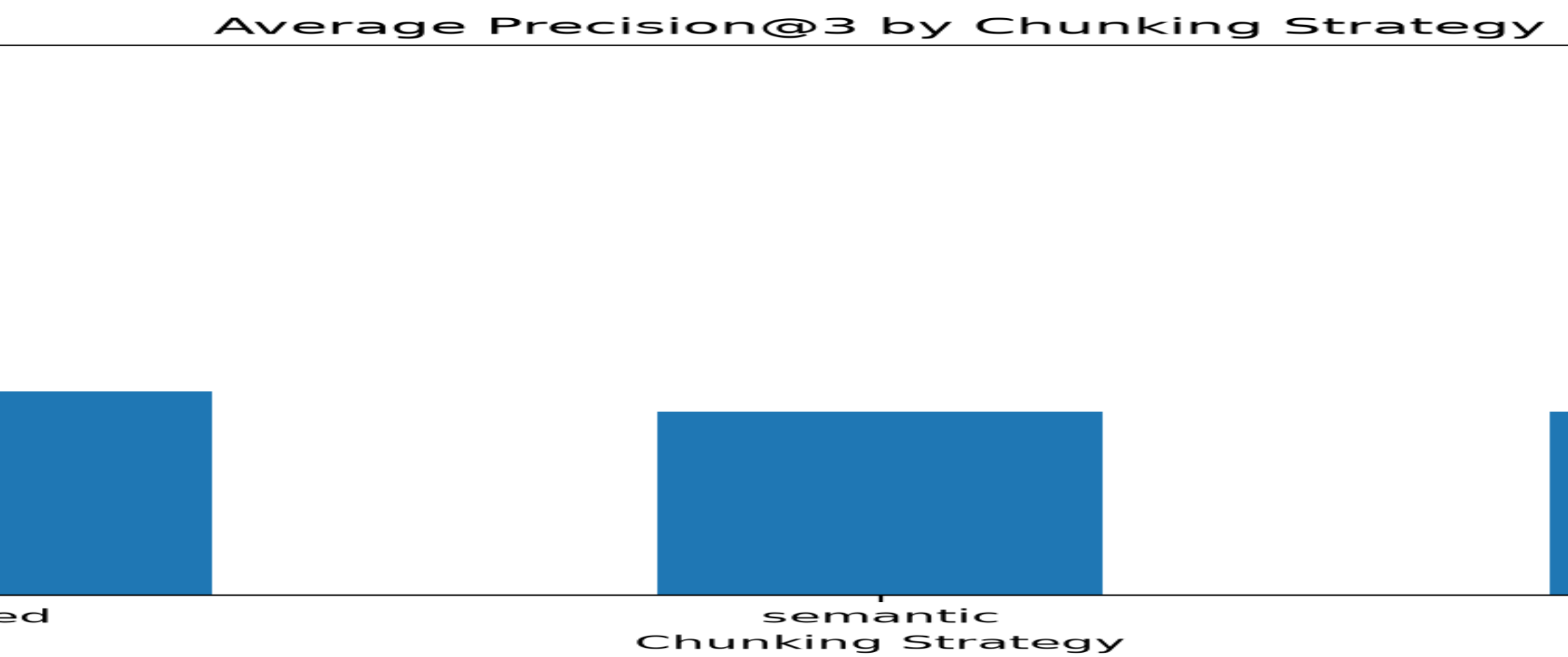
Chunk words	Precision@3	Recall@3	MRR	Latency (ms)	Indexing (ms)
64.0	0.3704	1.0000	1.0000	81.79	94.62
128.0	0.3333	1.0000	1.0000	82.23	102.56
256.0	0.3333	1.0000	1.0000	80.13	98.98

5. Visual Analysis

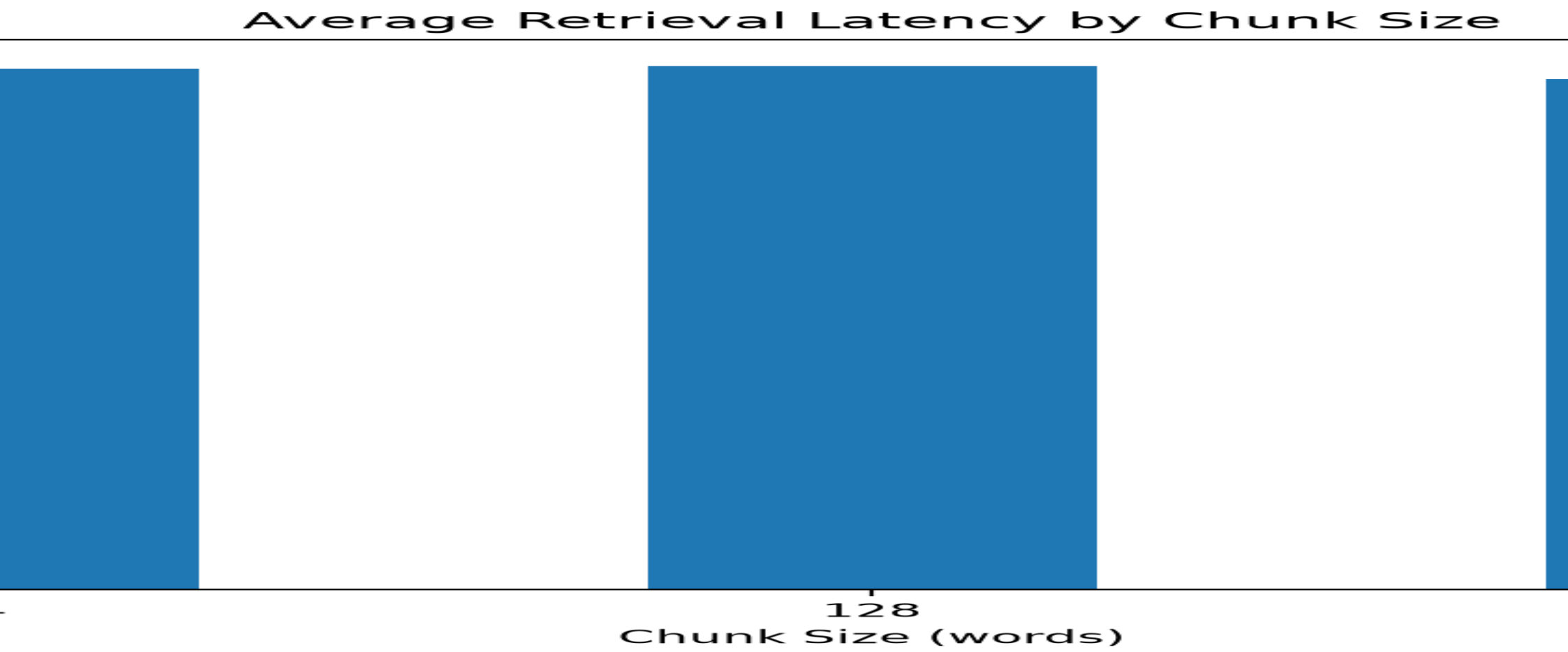
5.1 Average retrieval latency by embedding model



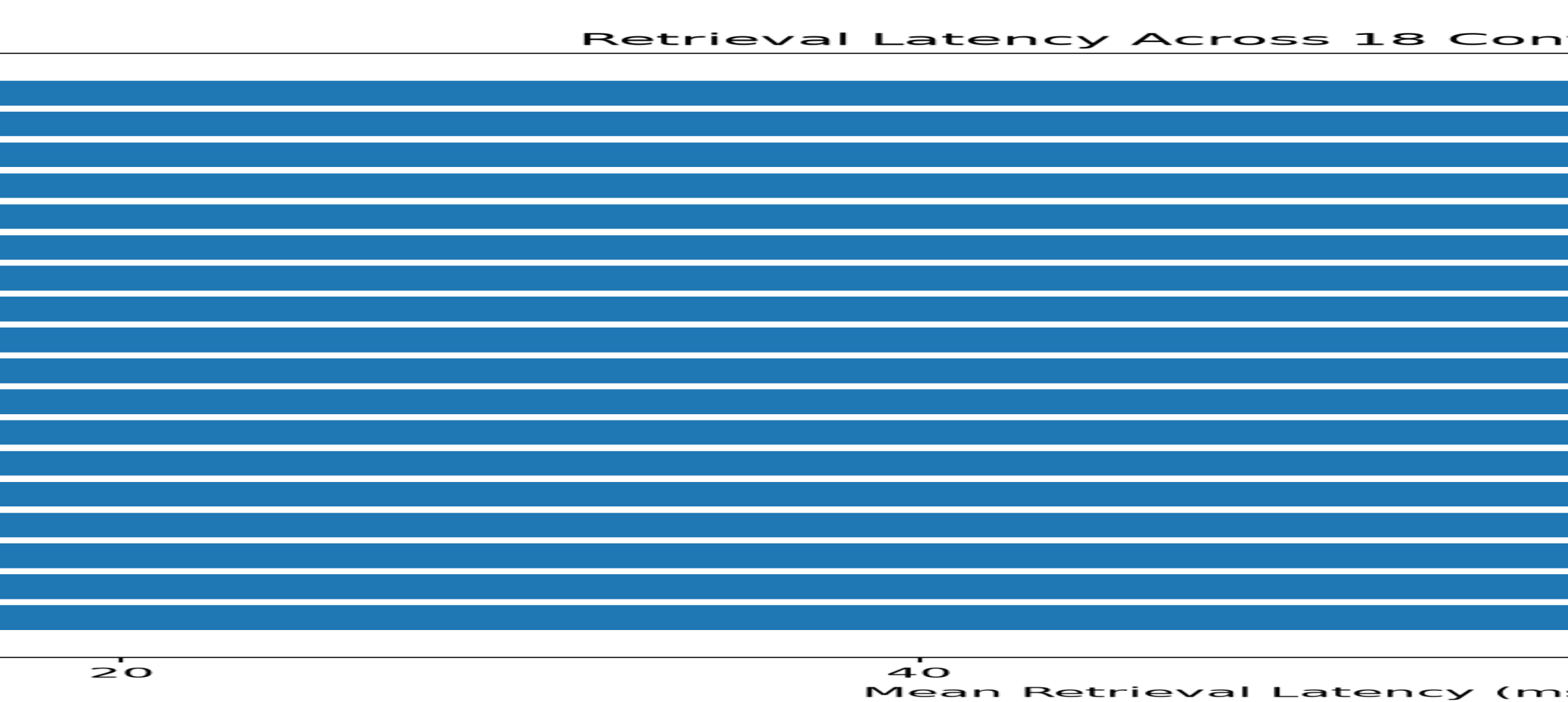
5.2 Average Precision@3 by chunking strategy


Average Precision@3 by Chunking Strategy
ed
semantic
Chunking Strategy

5.3 Average retrieval latency by chunk size



5.4 Retrieval latency across all 18 configurations


Retrieval Latency Across 18 Con
20
40
Mean Retrieval Latency (m

6. Configuration Ranking

The ranking below follows the benchmark's recorded optimizer score. The optimizer score combines retrieval accuracy and normalized speed.

Rank	Model	Strategy	Words	P@3	R@3	MRR	Latency ms	Index ms	Optimizer
1	minilm	sentence	128	0.333	1.000	1.000	75.62	81.31	0.9067
2	minilm	fixed	128	0.333	1.000	1.000	76.57	73.03	0.8886
3	minilm	sentence	256	0.333	1.000	1.000	77.11	72.14	0.8784
4	minilm	sentence	64	0.333	1.000	1.000	77.57	70.57	0.8695
5	minilm	semantic	256	0.333	1.000	1.000	77.91	118.33	0.8631
6	bge-small	fixed	64	0.444	1.000	1.000	79.47	70.99	0.8489
7	minilm	semantic	128	0.333	1.000	1.000	79.69	120.02	0.8290
8	bge-small	sentence	256	0.333	1.000	1.000	79.81	71.44	0.8268
9	bge-small	semantic	64	0.333	1.000	1.000	79.87	159.62	0.8257
10	minilm	fixed	256	0.333	1.000	1.000	80.63	76.73	0.8111
11	minilm	fixed	64	0.444	1.000	1.000	81.67	69.89	0.8068
12	bge-small	fixed	256	0.333	1.000	1.000	82.55	77.82	0.7747
13	bge-small	semantic	256	0.333	1.000	1.000	82.79	177.41	0.7701
14	bge-small	sentence	128	0.333	1.000	1.000	83.28	76.57	0.7606
15	minilm	semantic	64	0.333	1.000	1.000	85.31	119.97	0.7221
16	bge-small	sentence	64	0.333	1.000	1.000	86.84	76.66	0.6929
17	bge-small	fixed	128	0.333	1.000	1.000	86.85	76.11	0.6927
18	bge-small	semantic	128	0.333	1.000	1.000	91.36	188.32	0.6067

7. Best-ranked Configuration

Rank 1: minilm, sentence chunking, 128 words.

Precision@3: 0.3333; Recall@3: 1.0000; MRR: 1.0000; Mean retrieval latency: 75.62 ms; Indexing time: 81.31 ms; Optimizer score: 0.9067.

This is the best-ranked configuration under the benchmark's defined optimizer, not proof that it is universally best for other datasets.

8. Interpretation

All 18 configurations recorded Recall@3 and MRR of 1.0 in this small validation run. Precision@3 was generally 0.3333, with two configurations at 0.4444. These values indicate that the evaluated evidence was retrieved within the top three, while additional retrieved passages were not always relevant. The small number of answerable questions makes these metrics sensitive to individual examples.

MiniLM had a lower descriptive average retrieval latency than BGE-small in this run. Fixed chunking had the highest average Precision@3 among the three strategies, while semantic chunking had higher average indexing time. These are observations from this sample and configuration, not general claims about the models or chunking methods.

9. Methodology and Scoring

Optimizer retrieval accuracy = 0.45 × Recall + 0.35 × MRR + 0.20 × Precision. Speed score = 1 – (mean latency – cohort minimum) / (cohort maximum – cohort minimum). Balanced score = 0.70 × retrieval accuracy + 0.30 × speed score. Indexing time is reported separately and excluded from the speed score.

Precision uses the requested k as denominator; missing ranks count as non-relevant. Recall measures coverage of distinct validated evidence spans. MRR uses the first relevant passage for answerable questions. Unanswerable questions are assessed separately through abstention rate.

10. Limitations

• Only four validation questions were used, including three answerable questions.

• The benchmark used one repetition.

• The dataset is a fictional sample and is not representative of a real university knowledge base.

• Results are exploratory point estimates; no statistical significance is established.

• Answer generation was disabled; exact-match and groundedness were not evaluated.

• Retrieval performance does not establish generated-answer correctness.

• Hardware, caching and runtime conditions can affect latency.

11. Conclusion and Future Work

The benchmark completed all 18 configurations and recorded 72 measurements without failures. Under the current optimizer, MiniLM with Sentence chunking at 128 words ranked first. However, the small validation sample means the result should be treated as preliminary. Future evaluation should use a larger independently reviewed dataset, more repetitions, a held-out test set, and a separate evaluation of generated answer correctness and grounding.

12. Reproducibility References

- docs/verification/PES1UG24AM207_benchmark_export.json
- docs/evaluation/PES1UG24AM207_configuration_comparison.csv
- docs/evaluation/PES1UG24AM207_aggregate_analysis.md
- docs/evaluation/PES1UG24AM207_REPRODUCIBILITY_NOTES.md